

INTRODUCTION TO DOCKER IMAGES AND ITS NAMING

Q1. What is a Docker Image?

- A Docker image is a read-only template used to create Docker containers.
- It contains everything needed to run an application – **including**:
 - Application source code
 - Required libraries and dependencies
 - Environment variables
 - Configuration files

Key Characteristics:

- **Immutable**: Once built, it does not change.
- **Portable**: Can run on any system with Docker installed.
- **Layered**: Built from multiple layers using a Dockerfile.

❖ Common Docker image commands:

Command	Description
docker images	List local Docker images
docker pull <image>	Download an image from a registry
docker push <image>	Upload an image to a registry
docker login	Log in to a Docker registry
docker commit	Save container changes as a new image
docker tag	Rename or version an image
docker rmi or docker image rm	Remove an image from local storage
docker save	Save an image as a .tar archive
docker load	Load an image from a .tar archive
docker image prune	Delete unused images

[GENERAL INTERVIEW QUESTIONS]

Q1. What is a Docker image?

- A Docker image is a read-only template that contains everything needed to run an application, including the code, libraries, dependencies, and configuration. It is used to create Docker containers.

Q2. How is a Docker container different from a Docker image?

- A Docker image is a blueprint, whereas a container is a running instance of that image.

Q3. What is the naming convention of Docker images?

- *The format is:*
 - [registry/][username/]repository[:tag]
 - **Example**: docker.io/nginx:latest

Q4. What is Docker Hub?

- Docker Hub is a cloud-based repository where Docker users can store and share container images.

Q5. How do you log in to Docker Hub using the CLI?

- `docker login`

Q6. What does the `docker pull` command do?

- It downloads a Docker image from a remote registry like Docker Hub to the local system.

Q7. What does the `docker push` command do?

- It uploads a local Docker image to a remote registry.

Q8. What is the purpose of `docker tag`?

- It is used to create a new alias or version for a Docker image.

Q9. How do you remove a Docker image from the system?

- Using `docker rmi` or `docker image rm`.

Q10. What is the use of `docker save` and `docker load`?

- **`docker save`** is used to export an image to a .tar archive.
- **`docker load`** is used to import an image from a .tar archive.

Q11. What is the difference between `docker rmi` and `docker rm`?

- `docker rmi` removes images.
- `docker rm` removes containers.

Q12. What does the `docker image prune` command do?

- It removes all unused and dangling images to free up disk space

13. What's the difference between `docker image rm` and `docker image prune`?

- **`docker image rm`** removes a specified image.
- **`docker image prune`** removes all unused images automatically.